ENVS 193DS Final

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GitHub repo link

Homework set up

```
#read in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(ggplot2)
library(ggeffects)
library(flextable)
library(here)
library(MuMIn)
library(DHARMa)
library(rstatix)
#stored nest data as an object called nests
nests <- read_csv(here("data", "nest_data_final.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression, which is a generalized linear model with a binomial error distribution. In part 2, they used a chi-square test, which is used to identify if there is a relationship between two categorical variables.

b. Suggestions for rewriting

Part 1.

The distance to water edge does seem to predict the probability of American Avocet microhabitat use.

Part 2.

There does seem to be a relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further analysis

One additional variable that could influence the probability of American Avocet microhabitat use is the type of behavior they were displaying, which is a categorical variable. For the American Avocets, the researchers put behavior into four main categories: feeding (searching, probing, and dunking bills into the water), roosting (standing, sitting, preening), breeding (nest-building, incubating, brooding chicks, alarm calling, breeding display, copulation, or predator distraction display), and other (walking, swimming, or flushing). This variable might influence American Avocet habitat use as different behaviors can change the duration of how long they remain in a particular habitat.

Problem 2. Data analysis

a. Clean and wrangle

```
#created new object from nests
nests_clean <- nests |>
#cleaned column names
  clean_names() |>
#selected for tree height, ant_species, and case_control columns
  select(case_control, ant_species, height_cm) |>
#reordered names in ant_species column
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB",
                                   "RRB",
                                   "BBR")) |>
#filtered ant_species column to only have three required species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
#created new column with full scientific name for each ant species
  mutate(scientific_name = case_match(ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
```

```
    "BBR" ~ "Crematogaster nigriceps"))
#displayed structure of nests_clean data frame
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : Factor w/ 5 levels "AB","RRB","BBR",..: 2 2 2 2 1 2 3 2 1 3 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ scientific_name: chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaste
```

```
#displayed 10 random rows from the data frame but made it separate from all
 ↪  codes above by using separate pipe operator
nests_clean |>  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
   case_control ant_species height_cm scientific_name
          <dbl> <fct>           <dbl> <chr>
 1            0 RRB              210  Crematogaster mimosae
 2            1 RRB              238. Crematogaster mimosae
 3            0 RRB              185  Crematogaster mimosae
 4            0 RRB              118  Crematogaster mimosae
 5            0 RRB              150. Crematogaster mimosae
 6            1 RRB              195  Crematogaster mimosae
 7            0 BBR              105  Crematogaster nigriceps
 8            0 AB                85  Crematogaster sjostedti
 9            1 RRB              209  Crematogaster mimosae
10            1 RRB              247  Crematogaster mimosae
```

b. Response variable

For the case_control column, the 1s mean that a tree does contain a nest and the 0s mean that a tree does not contain a nest (is an “available” tree for comparison).

c. Purpose of study

Tree height is a continuous (numeric) variable and could influence the probability of nest occurrence as if tree height were to increase, the probability of nest occurrence would increase because there would most likely be less predators higher up on trees (Results section, paragraph 2). Acacia ant species is a categorical (nominal variable) and could influence the probability of nest occurrence as if a particular ant species at a nesting site was more aggressive, the probability of nest occurrence could increase as it would reduce nest predation risks (Discussion section, paragraph 1).

d. Table of models

Model number	Tree height	Ant species	Model description
0	X	X	All predictors (no interaction term, +)
1	X	X	All predictors (interaction term, *)

e. Run the models

```
#model 0: all predictors, no interaction term
nest_model0 <- glm(case_control ~ height_cm + scientific_name,
  data = nests_clean,
#response variable is binary so test should be logistic regression
  family = binomial)

#model 1: all predictors, interaction term
nest_model1 <- glm(case_control ~ height_cm * scientific_name,
  data = nests_clean,
#response variable is binary so test should be logistic regression
  family = binomial)
```

f. Select the best model

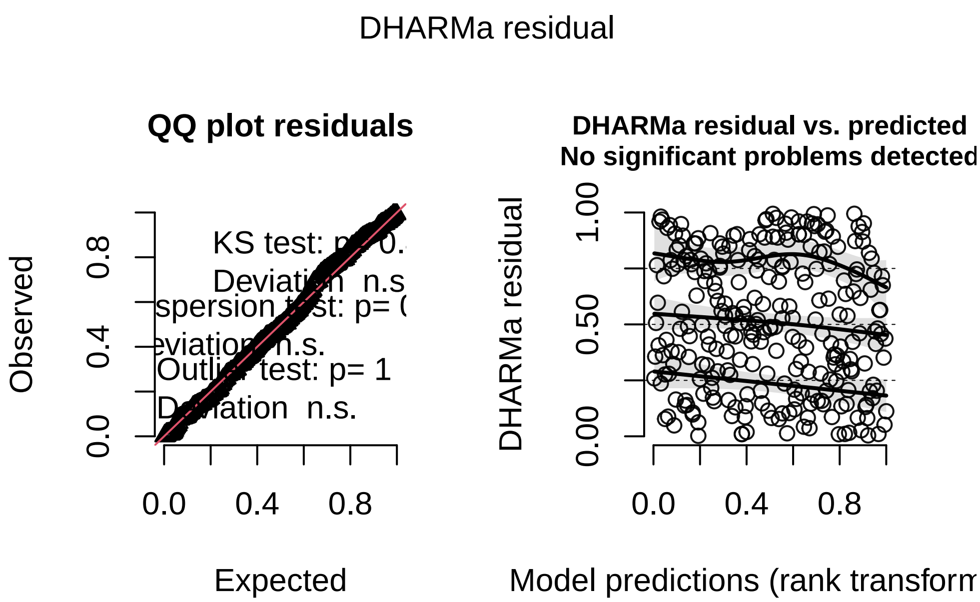
```
#calculated AIC for each model
AICc(
  nest_model0,
  nest_model1) |>
#arranged output in order of AIC
  arrange(AICc)
```

```
            df     AICc
nest_model1  6 238.1236
nest_model0  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) was the model where tree height (cm) and acacia ant species were interacting terms for the probability of nest occurrence.

g. Check the diagnostics

```
#checked diagnostics for model 1 using DHARMa residuals
plot(simulateResiduals(nest_model1))
```



Most of the points on the QQ plot follow the reference line, so we can assume the data is roughly normally distributed. We can only believe the observations collected were independent and the response and we already know the response variable will be distributed following a binomial distribution (0 or 1). For the plot on the right, since it says that there are no significant problems detected, we can also assume that there is a linear relationship between transformed expected response (nest occurrence probability) in terms of link function and explanatory variables (tree height and ant species). Therefore, all of the assumptions for a generalized linear model (GLM) are met.

h. Visualize the model predictions

i. Write a caption for your figure

j. Calculate model predictions

k. Interpret your results

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

b. Designing an analysis

c. Check any assumptions and run your analysis

d. Create a visualization

e. Write a caption

f. Write about your results

g. Sharing your affective visualization

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